But in the spring of 1995, with little warning, these modest hopes were wildly exceeded. Drawing on the work of a number of string theorists (including Chris Hull, Paul Townsend, Ashoke Sen, Michael Duff, John Schwarz, and many others), Edward Witten—who for two decades has been the world's most renowned string theorist—uncovered a hidden unity that tied all five string theories together. Witten showed that rather than being distinct, the five theories are actually just five different ways of mathematically analyzing a *single* theory. Much as the translations of a book into five different languages might seem, to a monolingual reader, to be five distinct texts, the five string formulations appeared distinct only because Witten had yet to write the dictionary for translating among them. But once revealed, the dictionary provided a convincing demonstration that—like a single master text from which five translations have been made—a single master theory links all five string formulations. The unifying master theory has tentatively been called *M-theory*, *M* being a tantalizing placeholder whose meaning—Master? Majestic? Mother? Magic? Mystery? Matrix?—awaits the outcome of a vigorous worldwide research effort now seeking to complete the new vision illuminated by Witten's powerful insight.

This revolutionary discovery was a gratifying leap forward. String theory, Witten demonstrated in one of the field's most prized papers (and in important follow-up work with Petr Hořava), *is* a single theory. No longer did string theorists have to qualify their candidate for the unified theory Einstein sought by adding, with a tinge of embarrassment, that the proposed unified framework lacked unity because it came in five different versions. How fitting, by contrast, for the farthest-reaching proposal for a unified theory to be, itself, the subject of a meta-unification. Through Witten's work, the unity embodied by each individual string theory was extended to the whole string framework.

Figure 13.1 sketches the status of the five string theories before and after Witten's discovery, and is a good summary image to keep in mind. It illustrates that M-theory is not a new approach, per se, but that, by clearing the clouds, it promises a more refined and complete formulation of physical law than is provided by any one of the individual string theories. M-theory links together and embraces equally all five string theories by showing that each is part of a grander theoretical synthesis.